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NATIONAL GOALS OF EDUCATION
Education in Kenya should:

- 1. Foster nationalism and patriotism and promote national unity.**
Kenya's people belong to different communities, cultures and regions, but these differences will not divide them. They must be able to live and interact as Kenyans. It is a paramount duty of education to help young people acquire this sense of nationalism by removing conflicts and promoting positive attitudes of respect and respect which enable them to live together in harmony and foster patriotism in order to make a positive contribution to the life of the nation.
- 2. Promote the social, economic, technological and industrial needs for national development.**
Education should provide the youth of the country with the capacity to be effective and innovative in the life of the nation.
 - a) Social Needs:**
Kenya's people belong to different communities, cultures and regions, but these differences will not divide them. They must be able to live and interact as Kenyans. It is a paramount duty of education to help young people acquire this sense of nationalism by removing conflicts and promoting positive attitudes of respect and respect which enable them to live together in harmony and foster patriotism in order to make a positive contribution to the life of the nation.
 - b) Economic Needs:**
Education in Kenya should produce citizens with the skills, knowledge, expertise and personal qualities that are required to support a growing economy. Kenya is building up a modern and independent economy which is in need of an adequate and relevant domestic workforce.
 - c) Technological and Industrial Needs:**
Education in Kenya should provide learners with the necessary skills and attitudes for industrial development. Kenya recognizes the rapid industrial and technological changes taking place, especially in the developed world. We can only be part of this development if our education system is deliberately focused on the knowledge, skills and attitudes that will prepare our young people for these changing global trends.
- 3. Promote individual development and self-fulfilment.**
Education should provide opportunities for the fullest development of individual talents and personality. It should help children to develop their potential as individuals and to appreciate the vital aspect of individual development in the building of a nation.
- 4. Promote sound moral and religious values.**
Education should provide for the development of knowledge, skills and attitudes that will enhance the acquisition of sound moral values and help children to grow up into self-disciplined, self-reliant and energetic individuals.

5. Promote social equity and responsibility.
Education should promote social equality and foster a sense of social responsibility within an education system which provides equal educational opportunities for all. It should give all children varied and challenging opportunities for collective activities and corporate social service irrespective of gender, ability or geographical environment.

6. Promote respect for and development of Kenya's rich and varied cultures.
Education should lead to the youth of Kenya a understanding of past and present cultures and their valid place in contemporary Kenya. Children should be able to blend the best of traditional culture with the changing requirements that will follow rapid development in order to build a stable and modern society.

7. Promote international consciousness and foster positive attitudes towards other nations.
Kenya is part of the international community. It is part of the global and interdependent network of peoples and nations. Education should therefore lead the youth of the country to accept membership of this international community with all the obligations and responsibilities that membership entails.

8. Promote positive attitudes towards good and ethical environmental protection.
Education should inculcate in young people the value of good health in order for them to avoid indulging in activities that will lead to physical or mental ill health. It should foster positive attitudes towards environmental development and conservation. It should lead the youth of Kenya to appreciate the need for a healthy environment.

TABLE 1: LIST OF SUBJECTS AT SENIOR SCHOOL			
Core Subjects	Elective Subjects	Science, Technology, Engineering & Mathematics (STEM)	
1. English 2. Kiswahili/SL 3. Core Mathematics 4. Community Service Learning (CSL)	5. Sports and Recreation 6. Music and Dance 7. Theatre and Film 8. Fine Arts 9. Literature in English 10. Indigenous Languages 11. Faith, Faith & Kiswahili 12. Sign Language 13. Arabic 14. French 15. Mandarin Chinese 16. Chinese Language 17. Hindi 18. Islamic Religious Education 19. Hindu Religious Education 20. Business Studies 21. History and Citizenship 22. Geography	23. Biology 24. Chemistry 25. Physics 26. General Science 27. Agriculture 28. Computer Studies 29. Home Science 30. Aviation 31. Health Science 32. Electricity 33. Metalwork 34. Power Mechanics 35. Woodwork 36. Textiles 37. Marine and Fisheries Technology	

Note:
a) Physical Education (PE) and Information Communication and Technology (ICT) will be offered to all learners to facilitate learning, enjoyment and acquisition of life skills.
b) Every school shall offer Pastoral Religious Programme of Instruction (PRPI) to enhance moral, spiritual and character development of the learners.
c) There shall be no fee for learners for personal or group study to deepen knowledge and enhance skills in pursuit of choice by the learners.

Table 2 provides guidelines on distribution of lessons at Senior School.

Core Subjects	Elective Subjects	Science, Technology, Engineering & Mathematics (STEM)
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By the end of Senior School, the learner should be able to:

1. Acquire scientific knowledge, skills, values and attitudes for everyday use, and further education and training.
2. Conduct a range of self-reliance, including collection of qualitative and quantitative data, analysis and interpretation of results.
3. Use scientific language in the study of Chemistry.
4. Develop skills and logical thinking in carrying out project work, field work excursions, where chemistry can be appreciated at a larger scale.
5. Relate Chemistry in the classroom to modern industrial processes and real world situations.
6. Apply critical thinking skills to explain phenomena and address challenges in day to day life.
7. Explain, manipulate, manage and conserve the environment for learning and sustainable development.
8. Apply principles of Chemistry and acquired skills to invent, formulate, construct, produce or manufacture useful products.
9. Demonstrate appreciation of diversity in people and cultures.
10. Manage pertinent and contemporary issues responsibly.

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Questions
1.0 Inorganic Chemistry	1.1 The Mole	1.1. The mole		
2.0 Physical Chemistry	2.1 Salts	2.1. Salts		
3.0 Organic Chemistry	3.1 Hydrocarbons	3.1. Hydrocarbons		

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Fractional distillation of crude oil a) Bonding in hydrocarbons. b) Homologous series c) Classification of hydrocarbons. (<i>Saturated and unsaturated hydrocarbons</i>) d) General molecular formula. e) Nomenclature of	a) identify sources of carbon compounds in nature b) classify hydrocarbons as saturated or unsaturated c) determine the molecular formulae of hydrocarbons using the general formula. d) assign names to molecules and their	a) review components of crude oil as obtained through fractional distillation. b) discuss the classifications of organic compounds as saturated or unsaturated. c) use the general formula of a homologous series to determine and write the molecular and structural formulae of hydrocarbons,	important in our lives?
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